

JAY KOTHARI

Pune · jaikothari10@gmail.com · (+91) 9428698017 · jaykothari.github.io

SUMMARY

Embedded Systems Engineer with a Master's degree in Engineering (specialization: Embedded Systems) with 11 years of work experience.

SKILLS

Languages:	C, C++
OS:	Deep understanding of operating system internals and implementation.
Linux:	Linux System Programming, Linux Kernel Programming, Linux Kernel Debugging.
Multi-Threading:	Multi-threaded application design and development.
RTOS:	Real-time driver development with Xenomai (RTDM).
QNX:	Hands-on experience with QNX RTOS and type-2 hypervisor.
DPDK:	Hands-on experience with DPDK development and maintenance.
Embedded Linux:	Porting Linux on custom boards; experience with Yocto build system.
ARM:	Worked on ARMv7/ARMv8 architecture.
Video Streaming:	Streaming Video/Audio over RTP/RTSP; raw TCP streaming.
Tools:	Git, Bootchartd, OpenIPMI, SQLite3, Qt, Live555.

EXPERIENCE

Incedo <i>Senior Technical Lead</i>	Pune, Maharashtra, India Oct 2022 – Present
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- Incedo is a product development partner with A10 Networks. Working in the system platform development team responsible for ACOS (Advanced Core Operating System).

Wipro <i>Senior Software Engineer</i>	Pune, Maharashtra, India March 2021 – Oct 2022
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- Telematics-platform bring-up and operating system services.

Visteon Corporation <i>Embedded Software Engineer</i>	Pune, Maharashtra, India July 2019 – Feb 2021
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- Smart Core is a single ECU that combines instrument cluster, head-up display (HUD), and Visteon's infotainment platform into a single domain controller. Worked on Smart Core platform bring-up.

Matrix Comsec <i>Senior Software Engineer</i>	Vadodara, Gujarat, India Sep 2015 – July 2019
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- Designed and developed Network Video Recorders(NVR) product core module.

SLTIET <i>Ad-hoc Lecturer, EC Department</i>	Rajkot, Gujarat, India June 2012 – June 2013
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PROJECTS

System Platform Team at Incedo / A10 Networks C, DPDK, Linux, Git

- New hardware platform bring-up and implementation of sensor monitoring (using OpenIPMI), watchdog, LED control, EEPROM; communication over I2C and IPMI (created raw IPMI library), getting information over the i2c for PSU.
- Userspace driver to communicate with proprietary FPGA hardware and communicate with SFP ports via i2c controller on FPGA.
- Implemented a user space driver for the SIO (Super I/O) chip using direct port I/O (iopl/inb/outb), enabling hardware monitoring and control (temperature, FAN speed, voltage, watchdog)
- Responsible for DPDK maintenance and problem solving.
- Implementation of Link Time Optimization (LTO) across ACOS.
- ARM-based platform from Ampere in ACOS; bring-up on vThunder and container environments.
- Designed a huge-page-based memory allocator for ACOS.

- RHEL 9 support for ACOS.
- Support for a high range of CPUs on ACOS; changes required in middleware.
- Maintenance of various platform image generation pipelines.
- Bug fixing and maintenance of middleware components.

Telematics Platform C, C++, Yocto, Embedded Linux, Git

Platform: Quectel SoC (AG35) and Renesas RH850 — a Telematics Control Unit (TCU) for which the team is responsible for BSP and OS services.

- Designed and developed a Persistence Manager using SQLite3 (client-server model).
- GPIO driver development for the Linux kernel.
- Implementation of low-power mode; handling synchronization between Vehicle Controller (VC) and SoC for service watcher mode, RTC sleep, superlock mode, and update mode.
- Defect resolution and product stabilization; analyzed and optimized boot time using Bootchartd.

Smart Core C, Android Linux Kernel, Git, Hypervisor, QNX

URL: visteon.com – SmartCore Cockpit Solution

Platform: Qualcomm SA6155/SA6145; Quectel module and Renesas RH850 on a single platform.

OS: QNX as type-2 hypervisor with Android as guest OS. Worked in OS/BSP team for porting and peripheral bring-up.

- Bring-up of UART and I2C peripherals (e.g., temperature sensor).
- Bring-up of TFT display and touch panel.
- Early display bring-up in QNX.
- Managed multiple product variants on a single codebase.
- Integrated Qualcomm SDK versions.

Network Video Recorder C, C++, Linux System Programming, Network Programming

URL: matrixvideosurveillance.com – NVR

Platform: HiSilicon Hi3536 (ARM Cortex-A17) and TI DM8168 (ARM Cortex-A8).

- Implementation of recording modules in the NVR product.
- Implemented RTSP Client and RTSP Server using the Live555 library.
- Designed and implemented the camera initiation module.
- Designed and implemented 4-channel synchronous playback.
- Implemented decode and display module for Hi3536 processor.
- Designed and implemented a hardware test application in Qt.
- Ported embedded Linux on the Hi3536 processor.

Real-Time Linux Driver (Xenomai RTDM) C, Linux Device Drivers

Code: github.com/JayKothari/i2c_rtdm

Article: opensourceforu.com – The Xenomai Project

Platform: BeagleBone Black with Cortex-A8 processor at 1 GHz.

Description: Design, implementation, and testing of Xenomai Real-Time drivers (RTDM) for I2C and UART controller on BeagleBone Black. This was the thesis project for the Master of Engineering degree.

EDUCATION

GTU PG School (Gujarat Technological University, Ahmedabad, Gujarat)

M.E. – VLSI & Embedded Systems

GPA: 8.1 / 10

June 2013 – June 2015

Atmiya University (Gujarat Technological University)

B.E. – Electronics and Communication Engineering

GPA: 8.2 / 10

June 2008 – June 2012

PERSONAL PROFILE

Date of Birth:	4th December 1989
Gender:	Male
Languages:	English, Hindi and Gujarati
Permanent Address:	1/7 Jairaj Plot, Palace Road, Rajkot – 360001, Gujarat, India
Nationality:	Indian

DECLARATION

I declare that the above information furnished by me is true to the best of my knowledge and I will present original documents whenever asked for.